ENGINEERING MATHEMATICS - II

UE20MA151

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Unit 4 : Inverse Laplace Transform

Session : 6

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1) Find the inverse Laplace transform of $\frac{s}{(s+4)^2}$

Solution:

$$L^{-1}\left[\frac{s}{(s+4)^2}\right] = L^{-1}\left[\frac{(s+4)-4}{(s+4)^2}\right]$$

$$= L^{-1}\left[\frac{1}{s+4} - \frac{4}{(s+4)^2}\right] = L^{-1}\left[\frac{1}{s+4}\right] - 4L^{-1}\left[\frac{1}{(s+4)^2}\right]$$

$$= e^{-4t} - 4t \cdot e^{-4t}$$

$$= e^{-4t}(1-4t)$$

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2) Find the inverse Laplace transform of $\dfrac{3s+2}{2s^2-4s+3}$

Solution:

Consider, $2s^2-4s+3 = 2\left(s^2-2s+1+\frac{1}{2}\right)$

$$= 2\left[(s-1)^2+\frac{1}{2}\right]$$

$$\therefore L^{-1}\left[\frac{3s+2}{2s^2-4s+3}\right] = L^{-1}\left[\frac{3s+2}{2\left\{(s-1)^2+\frac{1}{2}\right\}}\right]$$

$$= \frac{1}{2}L^{-1}\left[\frac{3s+2}{(s-1)^2+\frac{1}{2}}\right]$$

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$$L^{-1}\left[\frac{3s+2}{2s^2-4s+3}\right] = \frac{1}{2}L^{-1}\left[\frac{3(s-1)+5}{(s-1)^2+\frac{1}{2}}\right]$$

$$= \frac{3}{2}L^{-1}\left[\frac{s-1}{(s-1)^2+\left(\frac{1}{\sqrt{2}}\right)^2}\right] + \frac{5}{2}\sqrt{2}\cdot L^{-1}\left[\frac{\frac{1}{\sqrt{2}}}{(s-1)^2+\left(\frac{1}{\sqrt{2}}\right)^2}\right]$$

$$= \frac{3}{2}e^t\cdot\cos\left(\frac{t}{\sqrt{2}}\right) + \frac{5}{\sqrt{2}}e^t\sin\left(\frac{t}{\sqrt{2}}\right)$$

$$= \frac{e^t}{2}\left[3\cos\left(\frac{t}{\sqrt{2}}\right) + 5\sqrt{2}\sin\left(\frac{t}{\sqrt{2}}\right)\right]$$

3) Find the inverse Laplace transform of $\frac{2s^2-6s+5}{s^3-6s^2+11s-6}$.

Solution:

Consider $s^3-6s^2+11s-6 = (s-1)(s-2)(s+3)$

To find the inverse Laplace transform, first we shall resolve the given rational function into partial fractions.

Let $$\frac{2s^2-6s+5}{(s-1)(s-2)(s-3)} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s-3}$$

$$\Rightarrow 2s^2-6s+5 = A(s-2)(s-3) + B(s-1)(s-3) + C(s-1)(s-2)$$

Put $s=1$, $1 = 2A$ $\therefore A = \frac{1}{2}$

Put

$s=2, \quad 1=-B \quad \therefore B=-1$

$s=3, \quad 5=2C \quad \therefore C=\frac{5}{2}$

$$\therefore L^{-1}\left[\frac{2s^2-6s+5}{(s-1)(s-2)(s-3)}\right]=\frac{1}{2}L^{-1}\left[\frac{1}{s-1}\right]-L^{-1}\left[\frac{1}{s-2}\right]+\frac{5}{2}L^{-1}\left[\frac{1}{s-3}\right]$$

$$=\frac{1}{2}e^{t}-e^{2t}+\frac{5}{2}e^{3t}.$$

Thus,

$$L^{-1}\left[\frac{2s^2-6s+5}{s^3-6s^2+11s-6}\right]=\frac{1}{2}e^{t}-e^{2t}+\frac{5}{2}e^{3t}$$

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4) Find the inverse Laplace transform of $\log\left(1+\frac{1}{s^2}\right)$.

Solution:

Let $F(s) = \log\left(1+\frac{1}{s^2}\right) = \log\left(\frac{s^2+1}{s^2}\right)$

$$F(s) = \log(s^2+1) - \log s^2$$

$$F'(s) = \frac{2s}{s^2+1} - \frac{2s}{s^2}$$

$$-F'(s) = \frac{2}{s} - \frac{2s}{s^2+1}$$

$$L^{-1}\{-F'(s)\} = L^{-1}\left\{\frac{2}{s} - \frac{2s}{s^2+1}\right\}$$

$$t \cdot f(t) = 2 - 2\cos t \quad \left(\because L^{-1}\{-F'(s)\} = t \cdot f(t) \right)$$

$$f(t) = \frac{2(1-\cos t)}{t}$$

5) Find the inverse Laplace transform of $\frac{e^{-s} - 3e^{-s}}{s^2}$.

Solution:

$$L^{-1}\left\{\frac{e^{-s} - 3e^{-s}}{s^2}\right\} = L^{-1}\left\{\frac{e^{-s}}{s^2}\right\} - 3L^{-1}\left\{\frac{e^{-3s}}{s^2}\right\}$$

We know that $L^{-1}\{e^{-as} \cdot F(s)\} = f(t-a) \cdot u(t-a)$

$$= (t-1)\,u(t-1) - 3 \cdot (t-3)\,u(t-3).$$

THANK YOU

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